AS · Cambridge (CIE) · Physics

34 mins 34 questions

Multiple Choice Questions

Linear Momentum & its Conservation

Conservation of Momentum / Elastic & Inelastic Collisions

Easy (10 questions)	/10
Medium (13 questions)	/13
Hard (11 questions)	/11
Total Marks	**/34**

Scan here to return to the course
or visit savemyexams.com

© 2026 Save My Exams, Ltd.

Easy Questions

1 What is the principle of conservation of momentum?

A. force is equal to the rate of change of momentum

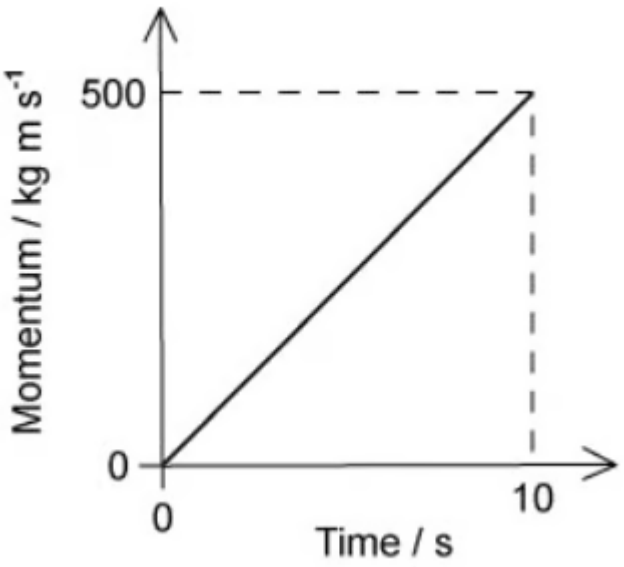
B. momentum is the product of mass and velocity

C. the total momentum of two bodies after collision is equal to their total momentum before collision

D. the total momentum of a system remains constant provided no external force acts on it

(1 mark)

2 The graph shows how the momentum of a motorcycle changes with time.



What is the resultant force on the motorcycle?

A. 50 N

B. 500 N

C. 2500 N

D. 5000 N

(1 mark)

© 2026 Save My Exams, Ltd.

3 Two bodies travelling in a straight line collide in a perfectly elastic collision. Which of the following statements must be correct?

A. the initial speed of one body will be the same as the final speed of the other body

B. the relative speed of approach between the two bodies equals their relative speed of separation

C. the total momentum is conserved but the total kinetic energy will be reduced

D. one of the bodies will be stationary at one instant

(1 mark)

4 Which row correctly states whether momentum and kinetic energy are conserved in an inelastic collision in which there are no external forces?

	Momentum	**Kinetic energy**
A	conserved	conserved
B	not conserved	conserved
C	conserved	not conserved
D	not conserved	not conserved

(1 mark)

5 Which quantity has the same base units as momentum?

A. density × volume × velocity

B. density × energy

C. pressure × area

D. weight ÷ area

(1 mark)

© 2026 Save My Exams, Ltd.

6 Which of the following is a statement of the principle of conservation of momentum?

A. in an elastic collision momentum is constant

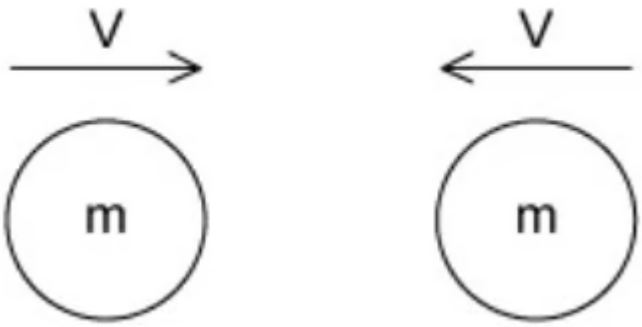
B. momentum is the product of mass and velocity

C. the force acting on a body is proportional to its rate of change of momentum

D. the momentum of an isolated system is constant

(1 mark)

7 Two similar spheres, each of mass *m* and travelling with speed *v*, are moving towards each other



The spheres have a head-on elastic collision

Which statement is correct?

A. the spheres stick together on impact

B. the total kinetic energy after impact is mv^2

C. the total kinetic energy before impact is zero

D. the total momentum before impact is $2mv$

(1 mark)

© 2026 Save My Exams, Ltd.

8 Trolley X, moving along a horizontal frictionless track, collides with a stationary trolley Y. The two trolleys become attached and move off together.

Which statement about this interaction is correct?

A. some of the kinetic energy of trolley X is changed to momentum in the collision

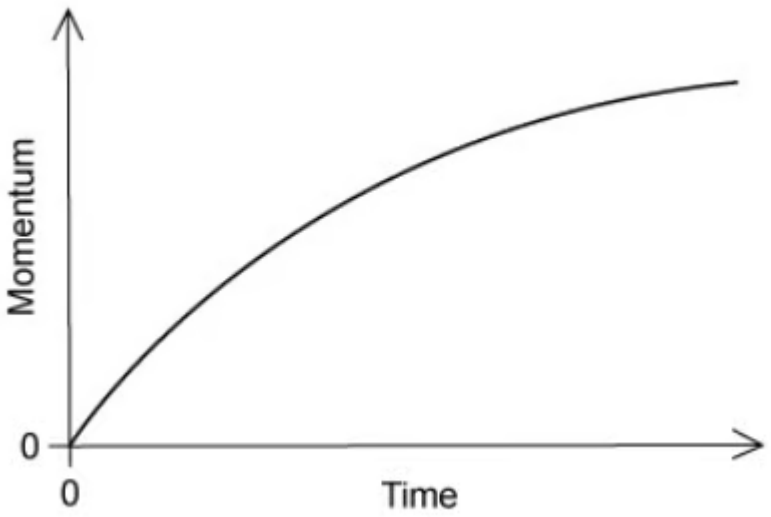
B. some of the momentum of trolley X is changed to kinetic energy in the collision

C. trolley X loses some of its momentum as heat in the collision

D. trolley X shares its momentum with trolley Y but some kinetic energy is lost

(1 mark)

9 A car accelerates from rest. The graph shows the momentum of the car plotted against time.



What is the meaning of the gradient of the graph at a particular time?

A. the velocity of the car

B. the resultant force on the car

C. the kinetic energy of the car

D. the rate of change of kinetic energy of the car

(1 mark)

© 2026 Save My Exams, Ltd.

10 Which statement about a ball that strikes a tennis racket and rebounds is **always** correct?

A. total kinetic energy of the ball is conserved

B. total kinetic energy of the system is conserved

C. total momentum of the ball is conserved

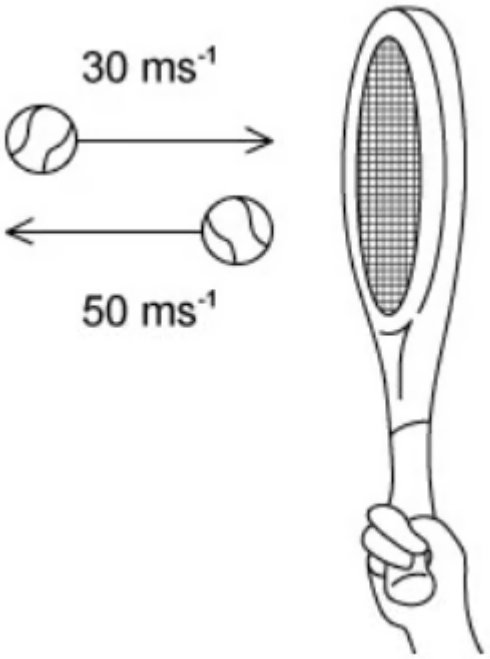
D. total momentum of the system is conserved

(1 mark)

© 2026 Save My Exams, Ltd.

Medium Questions

1 A tennis ball of mass 60 g is struck by a tennis racket. The velocity of the ball changes as shown



What is the magnitude of the change in momentum of the ball?

A. $1.2\ \text{kg m s}^{-1}$

B. $4.8\ \text{kg m s}^{-1}$

C. $1200\ \text{kg m s}^{-1}$

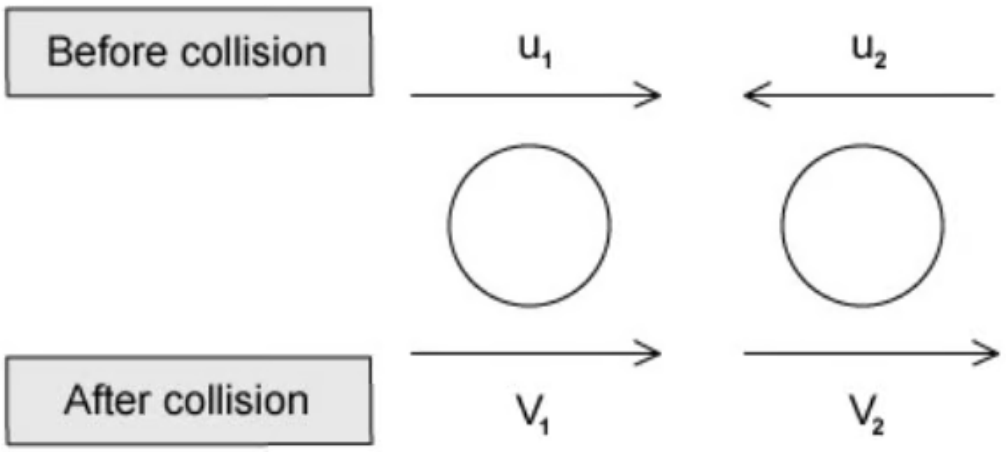
D. $4800\ \text{kg m s}^{-1}$

(1 mark)

2 Two spheres approach each other along the same straight line. Their speeds are u_1 and u_2

© 2026 Save My Exams, Ltd. | Get more and ace your exams at savemyexams.com

before collision, and v_1 and v_2 after collision, in the directions shown below



Which equation is correct if the collision is perfectly elastic?

A. $u_1 - u_2 = v_2 + v_1$

B. $u_1 - u_2 = v_2 - v_1$

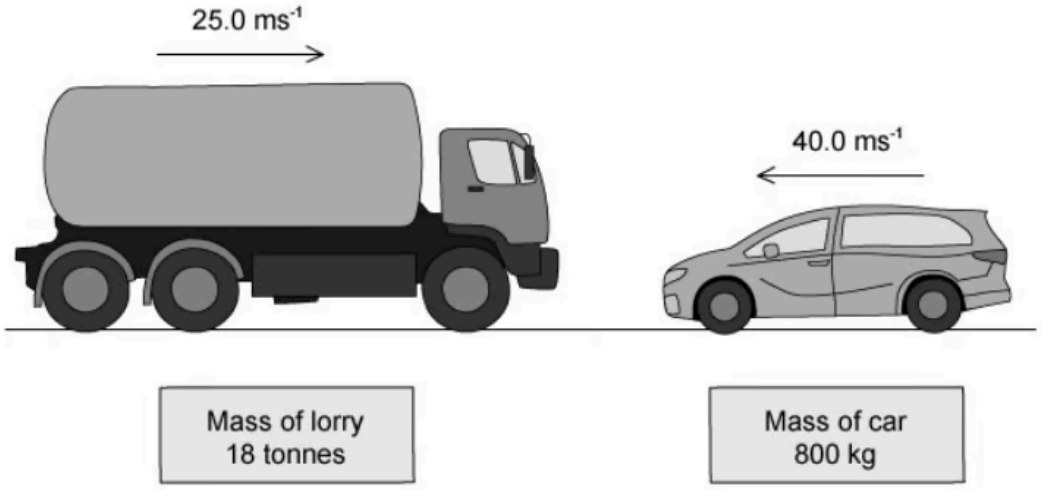
C. $u_1 + u_2 = v_2 - v_1$

D. $u_1 + u_2 = v_2 + v_1$

(1 mark)

© 2026 Save My Exams, Ltd.

3 A lorry of mass 18 tones is travelling at 25.0 $m\,s^{-1}$



A car of mass 800 kg is travelling at 40.0 $m\,s^{-1}$ towards the lorry.

What is the magnitude of the total momentum?

A. 282 kN s

B. 418 kN s

C. 482 kN s

D. 1222 kN s

(1 mark)

4 The diagram shows the masses and velocities of two trolleys about to collide



After the impact they move off together

What is the total kinetic energy of the trolleys after the collision?

A. 6.0 J

B. 3.7 J

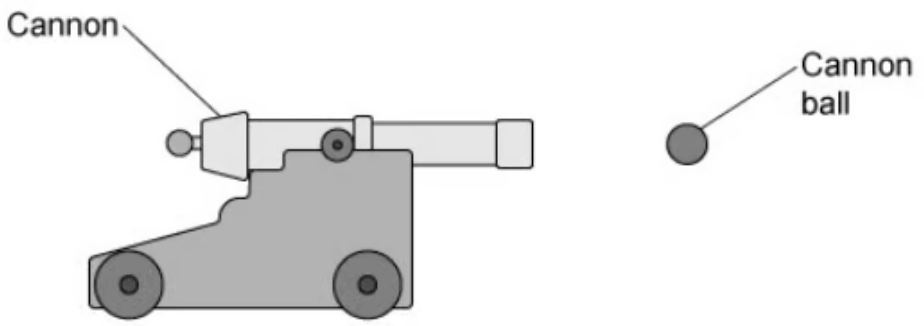
C. 0.35 J

D. 0.02 J

(1 mark)

5 The diagram shows a cannon ball fired from a cannon

The mass of the cannon is 1600 kg and the mass of the cannon ball is 20 kg



The recoil velocity of the cannon is 5 $\mathrm{m\,s^{-1}}$ horizontally

What is the horizontal velocity of the cannon ball?

A. 100 $\mathrm{m\,s^{-1}}$

B. 320 $\mathrm{m\,s^{-1}}$

C. 400 $\mathrm{m\,s^{-1}}$

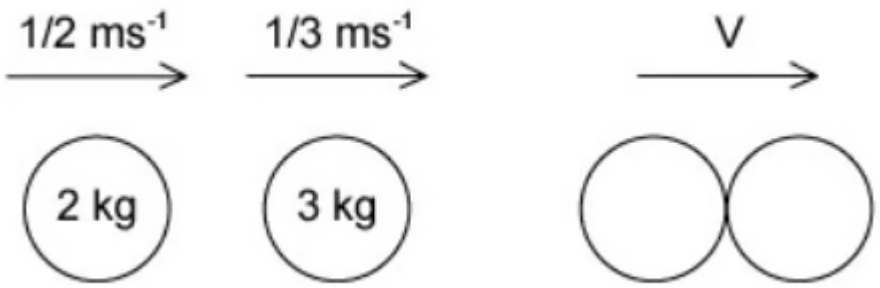
D. 8000 $\mathrm{m\,s^{-1}}$

(1 mark)

6 A ball of mass 2 kg travelling at ½ $\mathrm{m\,s^{-1}}$ strikes a ball of mass 3 kg travelling at ⅓ $\mathrm{m\,s^{-1}}$.

© 2026 Save My Exams, Ltd.

Both balls are moving along the same straight line as shown



After collision, both balls move at the same velocity *v*

What is the magnitude of the velocity *v*?

A. $\frac{1}{5}\ \text{m s}^{-1}$

B. $\frac{2}{5}\ \text{m s}^{-1}$

C. $\frac{3}{2}\ \text{m s}^{-1}$

D. $\frac{5}{2}\ \text{m s}^{-1}$

(1 mark)

7 A molecule of mass *m* travelling at speed *v* hits a wall in a direction perpendicular to the wall. The collision is elastic.

What are the changes in the momentum and in the kinetic energy of the molecule caused by the collision?

© 2026 Save My Exams, Ltd. | Get more and ace your exams at savemyexams.com

	Change in momentum	Change in kinetic energy
A	0	0
B	mv	mv^2
C	mv^2	0
D	$2mv$	0

(1 mark)

8 A moving object strikes a stationary object. The collision is inelastic. The objects move off together.

Which row shows the possible values of total momentum and total kinetic energy for the system before and after the collision?

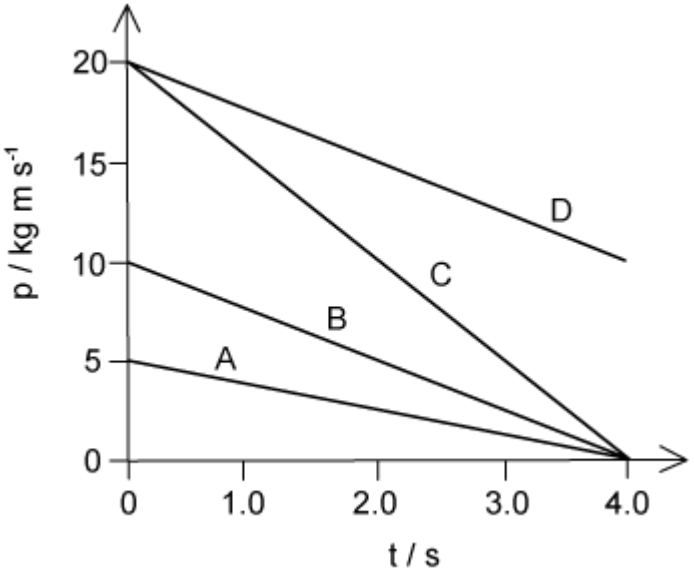
	Total momentum before collision / kg m s^{-1}	Total momentum after collision / kg m s^{-1}	Total kinetic energy before collision / J	Total kinetic energy after collision / J
A	6	6	90	30
B	6	6	30	90
C	6	2	90	30
D	6	6	90	90

(1 mark)

9 A resultant force of 5 N acts on a body for a time of 4.0 s

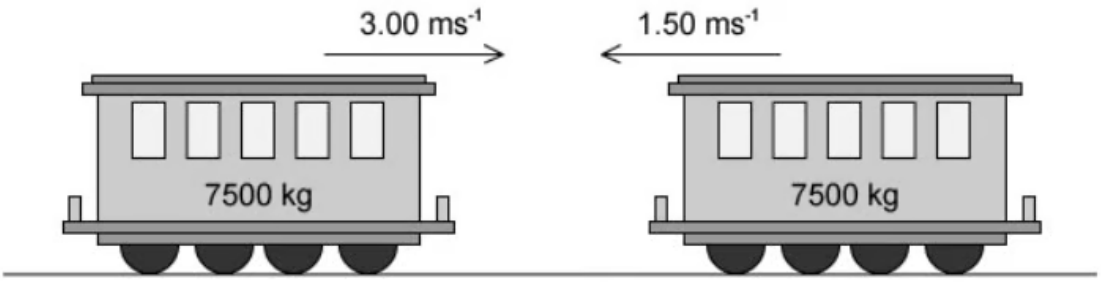
© 2026 Save My Exams, Ltd. | Get more and ace your exams at savemyexams.com

Which graph could show the variation with time *t* of the momentum *p* of the body?



(1 mark)

10 Two train carriages each of mass 7500 kg roll toward one another on a level track. One is travelling at 3.00 m s^{-1} and the other at 1.50 m s^{-1} as shown.



They collide and join together.

What is the kinetic energy lost during the collision?

A. 4200 J

B. 7500 J

C. 11 250 J

D. 38 000 J

(1 mark)

11 A spacecraft of mass 1000 kg and a fuel module of mass m are initially travelling together at 13 m s^{-1} in a straight line.

The fuel module separates from the rocket and continues to travel along the same line at 5 m s^{-1} while the spacecraft continues to travel along the same line at 17 m s^{-1}.

What is the mass m of the fuel module?

A. 300 kg

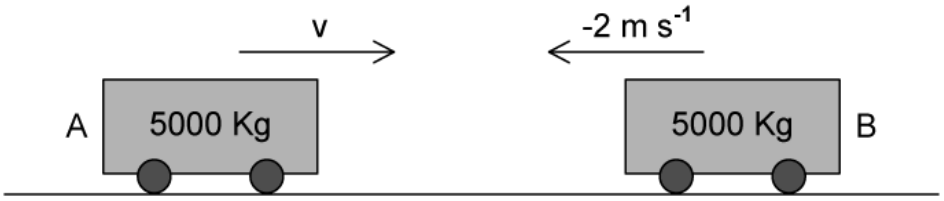
B. 500 kg

C. 800 kg

D. 1000 kg

(1 mark)

12 Two trolleys, A and B, each have a mass of 5000 kg. Trolley A travels at speed v towards trolley B. Trolley B travels at 2.0 m s^{-1} in the opposite direction, as shown.



The trolleys collide and join together with a combined kinetic energy of 1250 J.

What is the velocity of trolley A before the collision?

A. −1.0 m s^{-1}

B. 3.0 m s^{-1}

C. 3.4 m s^{-1}

D. 4.0 m s^{-1}

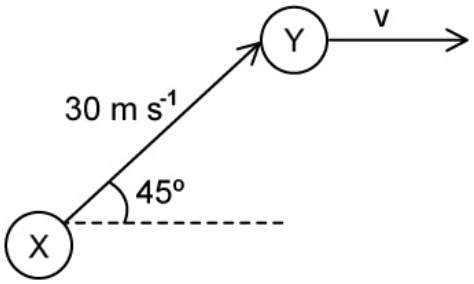
(1 mark)

13 Two identical billiard balls of mass m are placed on a smooth horizontal surface.

© 2026 Save My Exams, Ltd. | Get more and ace your exams at savemyexams.com

Ball Y is initially at rest. Ball X approaches ball Y with a speed of 30 m s^{-1} at an angle of 45° to the direction shown.

The balls collide perfectly elastically. After the collision, ball Y moves with speed v, and ball X moves at right angles to the direction of motion of ball Y.



What is the speed v of ball Y after the collision?

A. 21 m s^{-1}

B. 26 m s^{-1}

C. 30 m s^{-1}

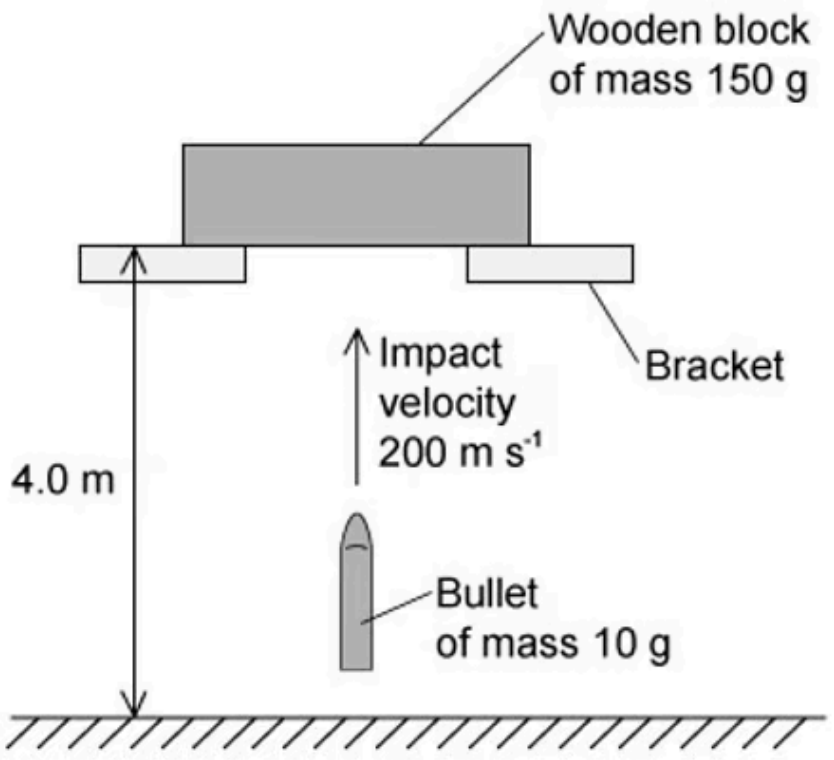
D. 39 m s^{-1}

(1 mark)

© 2026 Save My Exams, Ltd. | Get more and ace your exams at savemyexams.com

Hard Questions

1 A wooden block is freely supported on brackets at a height of 4.0 m above the ground, as shown



A bullet of mass 10.0 g is shot vertically upwards into the wooden block of mass 150 g. It embeds itself in the block. The impact causes the block to rise above its supporting brackets.

The bullet hits the block with a velocity of 200 m s^{-1}. How far above the ground will the block be at the maximum height of its path?

A. 8.0 m

B. 10.0 m

C. 12.0 m

D. 14.0 m

(1 mark)

© 2026 Save My Exams, Ltd. | Get more and ace your exams at savemyexams.com

2 Two railway trucks of masses *m* and 5 *m* move towards each other in opposite directions with speeds 3 *u* and *u* respectively. These trucks collide and stick together.

What is the speed of the trucks after the collision?

A. $\frac{u}{6}$

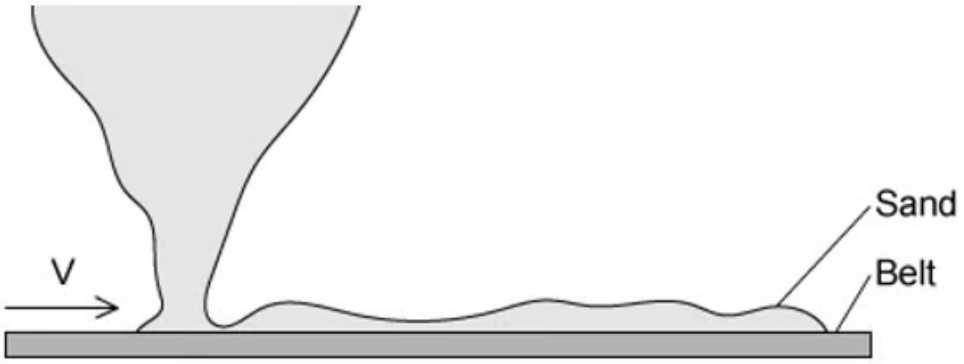
B. $\frac{3u}{6}$

C. $\frac{u}{3}$

D. $\frac{5u}{3}$

(1 mark)

3 Sand falls vertically on a conveyor belt at a rate of *m* kg s^{-1}



In order to keep the belt moving at constant speed *v*, the horizontal force that must be exerted is equal to

A. *mv*

B. ½ *mv*

C. mv^2

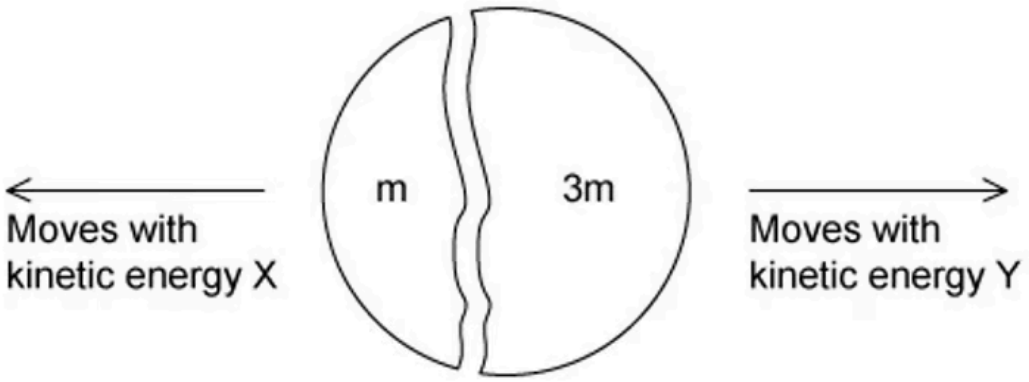
D. ½ mv^2

(1 mark)

© 2026 Save My Exams, Ltd. | Get more and ace your exams at savemyexams.com

4 A stationary body explodes into two components of masses *m* and 3 *m*.

The components gain kinetic energies X and Y respectively.



What is the value of the ratio $\frac{x}{y}$?

A. $\frac{6}{1}$

B. $\frac{3}{1}$

C. $\frac{1}{3}$

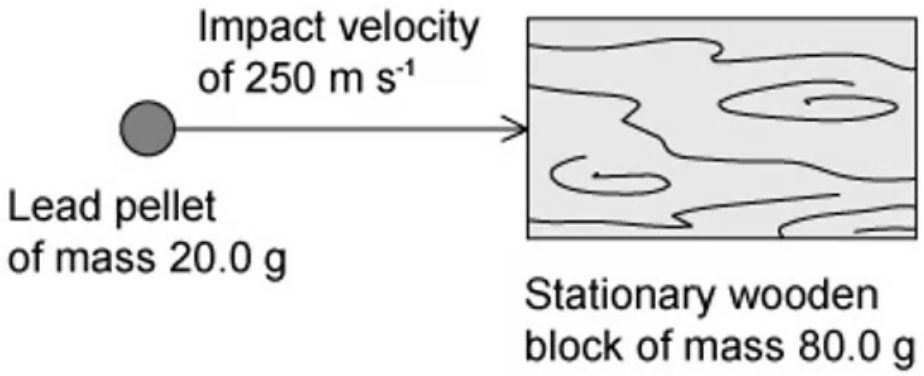
D. $\frac{1}{6}$

(1 mark)

5 A lead pellet of mass 20.0 g is shot horizontally into a stationary wooden block of mass 80.0 g. The pellet hits the block with an impact velocity of 250 m s^{-1}. It embeds itself in the block

© 2026 Save My Exams, Ltd. | Get more and ace your exams at savemyexams.com

and it does not emerge.



After the impact, the block and the pellet slide along a rough horizontal surface and come to rest in 8.4 s.

How far do the block and the pellet travel before coming to rest?

A. 175 m

B. 210 m

C. 260 m

D. 525 m

(1 mark)

6 A stationary nucleus has nucleon number *A*.

The nucleus decays by emitting a proton with speed *v* to form a new nucleus with speed *u*. The new nucleus and the proton move away from one another in opposite directions.

Which equation gives in terms of *A* and *u*?

A. $v = \left(\frac{A}{4} - 1\right) u$

B. $v = (A - 1)\, u$

C. $v = Au$

D. $v = (A + 1)\, u$

© 2026 Save My Exams, Ltd. | Get more and ace your exams at savemyexams.com

(1 mark)

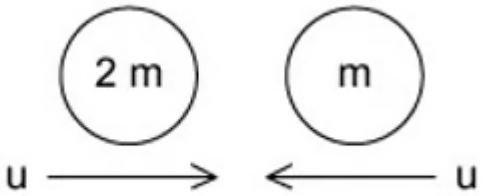
7 A particle X has speed *v* and collides with a stationary identical particle Y. The collision is perfectly elastic.

What are the speed and direction of motion of each of the two particles after the collision?

	X	Y
A	stationary	v to the right
B	$\frac{v}{2}$ to the right	$\frac{v}{2}$ to the right
C	$\frac{v}{2}$ to the left	$\frac{v}{2}$ to the right
D	*v* to the left	stationary

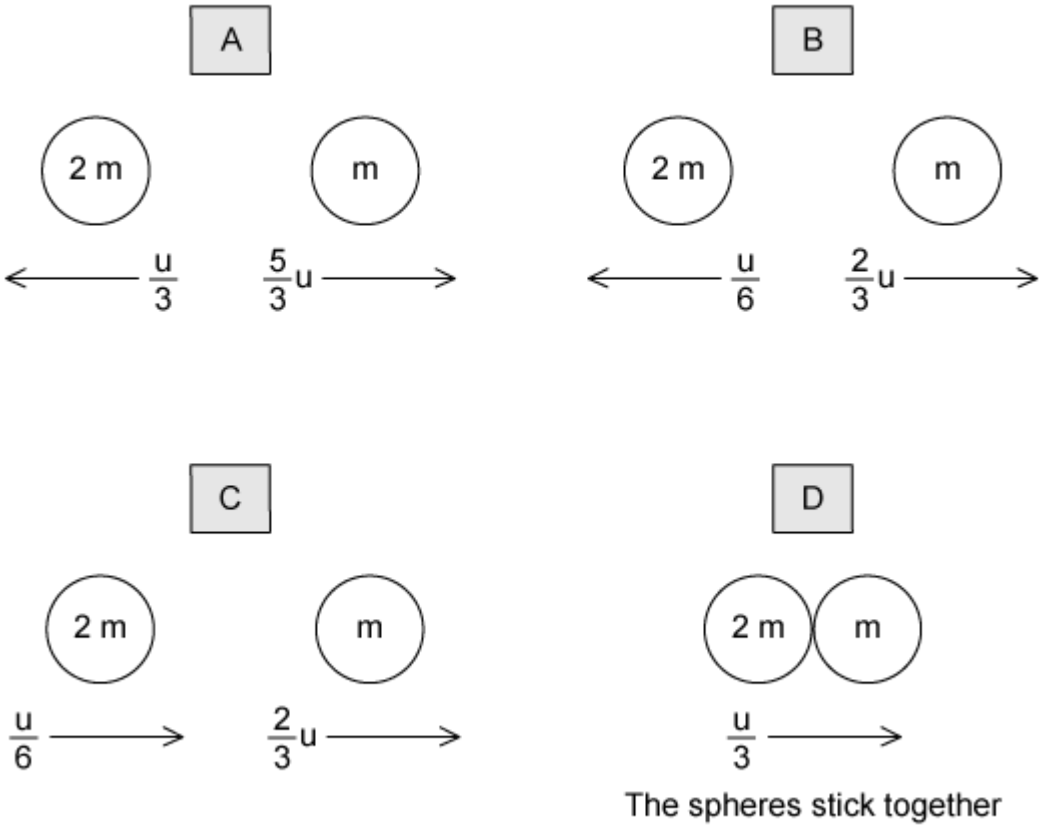
(1 mark)

8 The diagram shows two spherical masses approaching each other head-on at an equal speed *u*. One is of mass *m* and the other of mass 2 *m*.



Which diagram, showing the situation after the collision, is not consistent with the principle

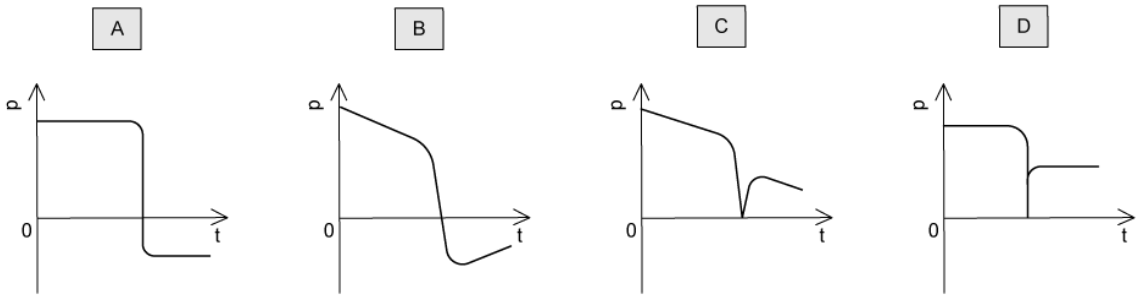
of conservation of momentum?



(1 mark)

9 An ice-hockey puck slides along a horizontal, frictionless ice-rink surface. It collides inelastically with a wall at right angles to its path, and then rebounds along its original path.

Which graph shows the variation with time *t* of the momentum *p* of the puck?



(1 mark)

10 A body, initially at rest, explodes into two masses M_1 and M_2 that move apart with speeds v_1 and v_2 respectively.

What is the ratio $\frac{v_1}{v_2}$?

A. $\frac{M_2}{M_1}$

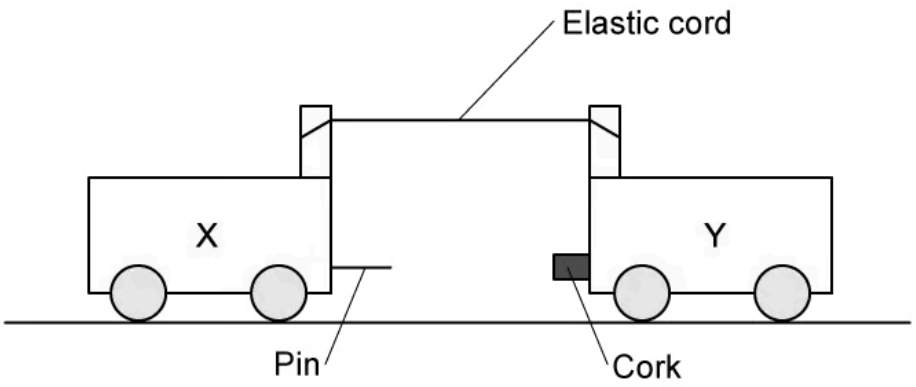
B. $\frac{M_1}{M_2}$

C. $\sqrt{\frac{M_2}{M_1}}$

D. $\sqrt{\frac{M_1}{M_2}}$

(1 mark)

11 Two trolleys, X and Y, are connected by an elastic cord, of negligible mass, as shown below. Trolley X is twice the mass of trolley Y.



The two trolleys are pulled apart so that the elastic cord is stretched. The trolleys are then released at the same time and move towards each other. The trolleys stick together on impact.

Just before the trolleys collide, the speed of trolley X is 20 cm s^{-1}.

© 2026 Save My Exams, Ltd.

What is the speed of the trolleys after they collide?

A. 0 cm s^{-1}

B. 10 cm s^{-1}

C. 20 cm s^{-1}

D. 30 cm s^{-1}

(1 mark)

© 2026 Save My Exams, Ltd.